Topic 6: Microbiology and Pathogens

This topic builds on knowledge of prokaryotes, eukaryotes, viruses and transport systems. It considers how some microorganisms act as pathogens. Details of how the human body responds to infection are included. The social, economic and ethical implications of the methods of treatment and control of the spread of infection are discussed. Microbial techniques are used in the isolation of bacteria and the investigation of the factors that affect their rate of growth.

Opportunities for developing mathematical skills within this topic include: recognising and making use of appropriate units in calculations, recognising and using expressions in decimal and standard form; using ratios, fractions and percentages; estimating results; finding and using power, exponential and logarithmic functions; using an appropriate number of significant figures; finding arithmetic means; using logarithms in relation to quantities that range over several orders of magnitude; translating information between graphical, numerical and algebraic forms; plotting two variables from experimental data; determining the intercept of a graph; drawing and using the slope of a tangent to a curve as a measure of rate of change. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:	
6.1	**Microbial techniques**
i	Understand the basic aseptic techniques used in culturing organisms.
ii	Understand the principles and techniques involved in culturing microorganisms.
iii	Understand the use of different media (broth cultures, agar and selective media).
iv	Understand the different methods of measuring the growth of a bacterial culture as illustrated by cell counts, dilution plating, mass and optical methods (turbidity).
v	Understand the different phases of a bacterial growth curve (lag phase, log phase, stationary phase and death phase) and calculate exponential growth rate constants.
CORE PRACTICAL 12: Investigate the rate of growth of bacteria in liquid culture taking into account the safe and ethical use of organisms.	
CORE PRACTICAL 13: Isolate individual species from a mixed culture of bacteria using streak plating taking into account the safe and ethical use of organisms.	